YUNHAI XIANG

Mathematician · Research Scientist · Software Engineer

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EDUCATION

Western University (University of Western Ontario) *2024 — Now*
Ph.D. (in progress) and M.Sc. in Mathematics *London, ON, Canada*
Advisor: Chris Hall
Specialization: Number Theory and Arithmetic Geometry

University of Waterloo *2018 — 2023*
B.Math. in Pure Mathematics, Honors, with Distinction *Waterloo, ON, Canada*

INDUSTRY EXPERIENCE

ByteDance 2019
Co-op Internship *Beijing, China*

· Supported data-informed product development for an education technology product
· Collected and reviewed educational data to support product analysis and application improvement

SKILLS

Colored by level of mastery: Expert, Advanced, Proficient, Familiar, Beginner

Languages	Mandarin, English, French, Japanese
Programming	Python, SQL, Lean, C/C++, JavaScript/TypeScript, C#, Rust, Go
DS/ML	NumPy, SciPy, Pandas, PyTorch, XGBoost, Sklearn
DevOps	Docker, Jenkins, FastAPI, Slurm, Kubernetes, Terraform
Graphics	Cinema 4D, Blender, Unity, Godot
Hobbies	Magic Tricks, Game Development, Jazz Music, Philosophy, Cooking

PUBLICATIONS & PREPRINTS

QEDBench: Quantifying the Alignment Gap in Automated Evaluation of University-Level Mathematical Proofs
Joint with Santiago Gonzalez, Alireza Amiri Bavandpour, Peter Ye, Edward Zhang, et al.
International Conference on Machine Learning (ICML) 2026

AWARDS

Mathematics Undergraduate Research Award 2022
University of Waterloo

· Highly competitive undergraduate research award of value $10000 at University of Waterloo. Participated in research project related to Weyl groups, supervized by Prof. Matthew Satriano.

CONFERENCES & SEMINARS

Conferences Attendance & Presentations
Attended academic conferences and delivered research or expository talks

· ∞-Type Café Summer School 2026, Online

- FrontierMath Open Problems Workshop, Summer 2026, UToronto
- Homotopy Theory, K-theory, and Topological Data Analysis Conference, Summer 2026, Western
- Using AI to Find Examples Workshop, under thematic program Universal Statistics in Number Theory, Winter 2026, Centre de Recherches Mathématiques, Université de Montréal
- University of Western Ontario Research in Computer Science (UWORCS), Winter 2026, Western
- *Rigid Analytic Geometry*, Ontario Graduate Mathematics Conference, Summer 2024, UWaterloo
- *Cohomology of Grassmannians*, Canadian Undergraduate Mathematics Conference, Summer 2022, Université Laval

Seminar Attendance & Presentations
Attended academic seminars and delivered research or expository talks

- *Hyperelliptic Jacobians and Hasse–Witt Matrix*, Algebraic Geometry Student Seminar, Western
- *Trace of Frobenius*, Winter 2026, Graduate Student Seminar, Western
- *Brauer Groups of Fields Part II*, Winter 2026, Algebraic Geometry Student Seminar, Western
- *Brauer Groups of Fields Part I*, Winter 2026, Algebraic Geometry Student Seminar, Western
- *Weyl Group and Root Datum*, Summer 2025, Algebraic Groups Seminar, Western
- *Hecke Operators*, Summer 2025, Algebraic Groups Seminar, Western
- *Parabolic and Borel Subgroups*, Summer 2025, Algebraic Groups Seminar, Western
- *Lie Algebras of Algebraic Groups*, Summer 2025, Algebraic Groups Seminar, Western
- *Admissible Topology*, Summer 2024, Rigid Analytic Geometry Seminar, Western
- *Non-Archimedean Fields*, Summer 2024, Rigid Analytic Geometry Seminar, Western
- *Cones and Affine Toric Varieties*, Summer 2024, Toric Varieties Learning Seminar, Western
- *An Easy Tour of Galois Cohomology*, Winter 2024, Graduate Student Seminar, Western
- *Introduction to Derived Categories*, Fall 2022, Algebraic Geometry Working Seminar, UWaterloo

TEACHING & MENTORING

Teaching Assistantship
Proctored and graded exams, graded homeworks, tutored at Help Centre, and taught tutorial sessions

- MATH 3150, *Elementary Number Theory I*, Fall 2024
- MATH 2155, *Mathematical Structures*, Fall 2025, Fall 2026
- CALCULUS 1500, *Calculus I for the Mathematical Sciences*, Fall 2026
- CALCULUS 1501, *Calculus II for Mathematical and Physical Sciences*, Winter 2025, Winter 2026, Summer 2026
- MATH 1229, *Methods of Matrix Algebra*, Summer 2024, Summer 2025
- MATH 1228, *Methods of Finite Mathematics*, Summer 2024, Summer 2025
- CALCULUS 1000, *Calculus I*, Winter 2024, Summer 2024, Winter 2025, Fall 2025, Winter 2026

Mathematical Outreach Lectures
Gave expository lectures to high school students or undergraduate students on advanced mathematical topics

- *Aperiodic Tilings*, Summer 2026, Western Summer Math Camp (Grade 11/12)
- *Surreal Numbers and the Game of Go*, Summer 2025, Western Summer Math Camp (Grade 10)
- *Mathematical Magic Tricks: Gilbreath's Principle*, Summer 2024, Western Summer Math Camp (Grade 11)
- *Why is There no Quintic Formula*, Summer 2024, Western Summer Math Camp (Grade 10)
- *Category Theory Demystified*, Winter 2023, Short Attention Span Math Seminar (Undergraduate)

Directed Reading Program Mentorship
Mentored undergraduate students on reading projects, culminating in a presentation

· Jude McNorgan Lubos, *Introduction to Quadratic Reciprocity*, Winter 2026

SELECTED SOFTWARE PROJECTS & OPEN SOURCE CONTRIBUTIONS

Games and Graphics Softwares

· Coming Soon.

Mathematics and Scientific Computing

· Coming Soon.

Data Science and Quantitative Finance

· Coming Soon.

CERTIFICATIONS

Language Proficiency

· Coming Soon.

Boot Camps & Programs

· Coming Soon.

Technical Certifications

· Coming Soon.

SERVICE AND VOLUNTEERING

Seminar and Conference Organization

· Graduate Student Research Seminar, Fall 2026, co-organized with Alex Zwart
· Algebraic Geometry Student Seminar, Summer 2026
· Algebraic Geometry Student Seminar, Winter 2026, co-organized with Alex Zwart
· Algebraic Geometry Student Seminar, Fall 2024, co-organized with Alex Zwart
· Toric Varieties Learning Seminar, Summer 2024, co-organized with Nikolas Rieke